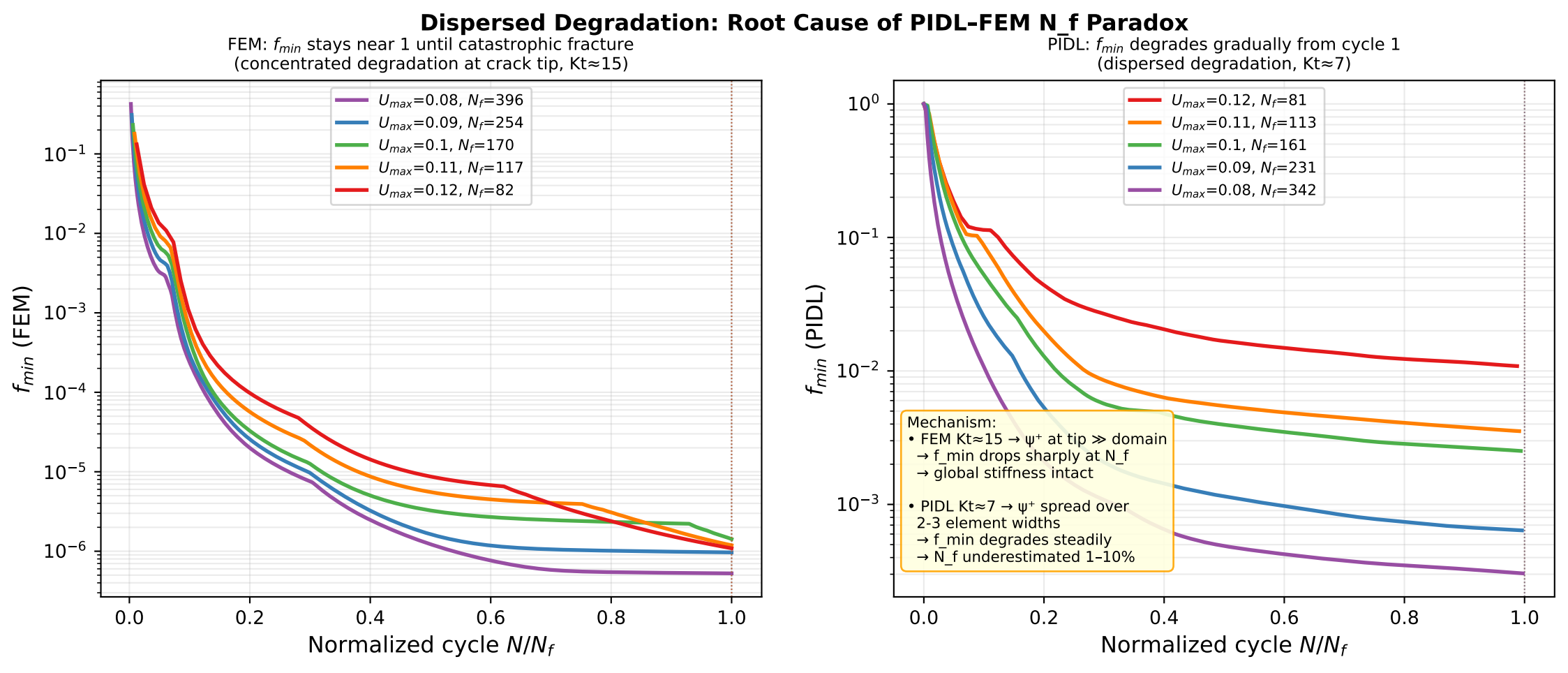
Dispersed Degradation: Root Cause of PIDL–FEM N_f Paradox
FEM: f_{min} stays near 1 until catastrophic fracture
(concentrated degradation at crack tip, Kt≈15)
PIDL: f_{min} degrades gradually from cycle 1
(dispersed degradation, Kt≈7)
U_{max}=0.08, N_f=396
U_{max}=0.09, N_f=254
U_{max}=0.1, N_f=170
U_{max}=0.11, N_f=117
U_{max}=0.12, N_f=82
U_{max}=0.12, N_f=81
U_{max}=0.11, N_f=113
U_{max}=0.1, N_f=161
U_{max}=0.09, N_f=231
U_{max}=0.08, N_f=342
f_{min} (FEM)
f_{min} (PIDL)
Normalized cycle N/N_f
Normalized cycle N/N_f
Mechanism:
• FEM Kt≈15 → ψ⁺ at tip ≫ domain
→ f_min drops sharply at N_f
→ global stiffness intact
• PIDL Kt≈7 → ψ⁺ spread over
2-3 element widths
→ f_min degrades steadily
→ N_f underestimated 1–10%